MILITARY SPECIFICATION

ARM SLEEVE, RADIOACTIVE CONTAMINANTS, PROTECTIVE

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for protective arm sleeves made of flexible vinyl plastic film.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

SPECIFICATIONS

FEDERAL

V-T-276	- Thread, Cotton
L-P-375	- Plastic Film, Flexible, Vinyl Chloride
JJ-W-155	- Webbing, Textile (Cotton, Elastic)
PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-T-43548	- Thread, Polyester, Cotton-covered
MIL-T-43624	- Thread, Polyester, Spun

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

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STANDARDS

FEDERAL

FED-STD-595 - Color

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.1 Other Government documents and publications The following other Government documents and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS

US POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402).

2.2 Other publications The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies shall be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies shall be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

2.3 Order of precedence In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Guide sample Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample, in which case this specification shall govern.

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3.2 First article When specified (see 6.2, 4.3, and 6.4) the contractor shall furnish a sample for first article inspection and approval.

3.3 Materials

3.3.1 Basic material The basic material shall be yellow plastic film, flexible, vinylchloride comforming to Type I, Class 2 of L-P-375, 0.0080 ± 0.0008 inch thick, except that the requirements for extraction in soapy water, volatility, resistance to weathering, cold crack, and lacquer lifting shall not apply, and no pinholes are allowed. The yellow color shall approximate the shade of color No. 23665 of FED-STD-595.

3.3.2 Webbing, Elastic The webbing shall be cotton elastic, unbleached or bleached white conforming to Type II, Class 1 of JJ-W-155 or a braided elastic webbing of equivalent grade may be used. The webbing shall be 5/16 (± 1/32) inch wide.

3.3.3 Thread The thread used for joining the ends of the elastic webbing shall be cotton, polyester or cotton/polyester conforming to V-T-276, MIL-T-43548 or MIL-T-43624. The size of the thread shall be suitable for the intended purpose. The color of the thread shall be natural or white.

3.3.4 Label Each arm sleeve shall be identified with the inscription specified below. The inscription shall be positioned on the upper part of the arm sleeve approximately 1 1/2 inches down from the folded edge of the armhole hem. The inscription shall be permanently printed, stamped or stenciled directly on the plastic or it may be printed on a pressure-sensitive label that will adhere satisfactorily to the plastic.

ARM SLEEVE, RADIOACTIVE CONTAMINANTS, PROTECTIVE
CONTRACT NO:
STOCK NO:

3.4 Design The sleeve, as shown in Figure 1, shall be made by electronically heat sealing a flat section of flexible vinyl plastic into a tube of the proper length and shape. It shall incorporate elastic webbings for drawing in around the wrist (see figure 1).

3.4.1 Figure The figure is furnished for information purposes only. When inconsistencies exist between the written specification and the figure, the written specification shall govern.

3.5 Patttern The pattern for the one piece arm sleeve shall be supplied by the contractor.

3.6 Heat sealing dies The contractor shall furnish the electronic heat sealing equipment and dies required to manufacture the arm sleeve.

3.7 Construction

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3.7.1 Replacement of damaged parts Care should be exercised during the cutting and manufacturing operations to assure that material defects and damages which may affect appearance or serviceability of the arm sleeve shall be excluded and replaced with non-defective material.

3.7.2 Manufacturing operation requirements The arm sleeve shall be manufactured in accordance with the following operations:

1. Cut the elastic webbing 8 (± 1/16) inches in length.

2. Overlap the cut ends of the elastic webbing 1/2 (± 1/16) inch and backstitch or bartack across the width of webbing, forming a continuous wrist loop approximately 7 inches in circumference.

3. Cut out the one piece arm sleeve from the specified plastic film in a size and shape that will meet the finished dimensions of Table I.

4. Overlap the two edges of the proposed length of arm seam 3/8 (± 1/16) inch and heat seal the entire length of seam 1/4 to 5/16 inch in width. The seam shall be of uniform width, continuous, free of bubbles, discoloration and have a minimum of distortion, wrinkling, and puckering.

5. Turn up the bottom or wrist edge of the arm sleeve approximately 1 1/4 inches to the inside of sleeve. Insert the elastic wrist loop and encase the loop with a continuous heat sealed seam 1/8 to 1/4 inch in width. The fused seam shall be located 7/8 to 1 inch from the bottom folded edge of arm sleeve.

6. Turn down the top or armhole edge of sleeve 5/8 (± 1/16) inch to the inside of sleeve and form a hem with a 1/8 to 1/4 inch wide continuous heat sealed seam located 1/4 to 5/16 inch from the folded edge of armhole.

7. Trim off any excess plastic to 1/8 inch or less from edge of seam without damaging seam.

8. Affix label (see 3.3.4).

3.8 Measurements The finished dimensions of the arm sleeve shall be as indicated in Table I and Figure 1. All measurements are in inches and shall be recorded with the arm sleeve laid out flat.

TABLE I - Finished dimensions

			Tolerance
(A)	Length of arm sleeve	18	± 1/4
(B)	Width of armhole	10	± 1/4
(C)	Width of wrist opening 1/		
	- with elastic relaxed	3 3/4	± 1/4
	- with elastic stretched to maximum	6 3/4	± 1/4
(D)	Width of upper arm 6 inches down from edge of armhole	9	± 1/4
(E)	Width of lower arm 4 inches up from edge of wrist	8	± 1/4

1/ Width of wrist opening before insertion of elastic webbing shall be 8 ± 1/4 inches.

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3.9 Workmanship The finished arm sleeves shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection The first article submitted in accordance with 3.2 shall be inspected for compliance with design, construction, workmanship, appearance and dimensional requirements.

4.4 Quality conformance inspection Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of reference specifications, drawings, and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 Examination of the end item Examination of the end item shall be in accordance with 4.4.2.1 and 4.4.2.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.4.3. The sample unit shall be one completely fabricated arm sleeve. The lot size shall be expressed in units of arm sleeves.

4.4.2.1 Visual examination The arm sleeve shall be examined for defects classified in accordance with the list below:

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Examine	Defects
Material Defects and Workmanship	1. Not material specified 2. Not color specified 3. Not uniform in color 4. Not smooth finish, i.e., any arm sleeve having a surface defect such as crazing, cracking, blistering, pitting, bubbling or permanent wrinkling 5. Any hole, cut, tear, rupture, or thin spot 6. Any repair patch or mend 7. Excess material adjacent to heat sealed seam not trimmed to 1/8 inch or less 8. Any embedded foreign material 9. Any open seam 10. Any inadequate heat seal, i.e., seam can be easily pulled apart with moderate hand pressure 11. Seam excessively puckered or not continuous 12. Any unspecified seam or heat sealed area 13. Width of heat sealed seam not as specified 14. Any permanent spot or stain 15. Label missing, misplaced, illegible or not as specified

4.4.2.2 Dimensional examination The arm sleeves shall be examined for dimensional defects. Any measurements deviating from dimensions and tolerances specified in 3.8 shall be scored as a defect.

4.4.2.3 Inspection level and acceptable quality levels The inspection level and acceptable quality levels (AQL's), expressed as defects per 100 units, for visual and dimensional examinations shall be as follows:

	Inspection level	AQL
For visual examination in 4.4.2.1	II	4.0
For dimensional examination in 4.4.2.2	S-3	4.0

4.4.3 Testing of the end item

4.4.3.1 Plasticizer exudation A test for exuded plasticizer shall be conducted on three finished arm sleeves randomly selected from each assembled shipping lot. Test shall be performed by wiping the surface of the arm sleeves using a clean dry white cloth, moderate hand pressure, and three back and forth wiping cycles. Staining of the cloth indicates that the plastic is unacceptable. Failure of any of the three armsleeves tested shall be cause for rejection of the shipping lot.

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4.5 Examination of packaging requirements An examination shall be made to determine that packaging, packing and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Defects of the closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units (DHU).

Examine	Defects
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, incorrect application of tape; bulging or distortion of container
Content	Number of arm sleeves per container more or less than specified 1/

1/ For this defect two shipping containers in the inspection sample shall be examined.

5. PACKAGING

5.1 Packaging Packaging shall be Level A, B or C as specified (see 6.2).

5.1.1 Level A Twenty-six (26) pairs of arm sleeves shall be laid out flat and neatly placed, one pair on top of another pair, in two equal stacks of thirteen (13) pairs each. The two stacks shall be reversed end to end in the shipping container so that the wrist end of one stack is oposite the armhole of the other stack. The two equal stacks shall be packed flat, side by side, in a snug fitting fiberboard box conforming to style RSC-L, type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. The fiberboard for the box liner shall conform to type CF, class domestic, variety SW or DW, grade 200 of PPP-B-636. Box closure shall be in accordance with the appendix of PPP-B-636.

5.1.2 Level B Twenty-six (26) pairs of arm sleeves shall be laid out flat and neatly placed, one pair on top of another pair, in two equal stacks of thirteen (13) pairs each. The two stacks shall be reversed end to end in the shipping container so that the wrist end of one stack is opposite the armhole

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of the other stack. The two stacks shall be packed flat, side-by-side, in a snug fitting fiberboard box conforming to style RSC-L, type CF, class domestic, variety DW, grade 200 of PPP-B-636. The fiberboard for the box liner shall conform to type CF, class domestic, variety SW or DW, grade 200 of PPP-B-636. Box closure shall be in accordance with the appendix of PPP-B-636.

5.1.3 Level C (Commercial) Arm sleeves shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the U.S. Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.2 Marking In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6 NOTES

6.1 Intended use Arm sleeves covered by this specification are intended to be used as protective garments by personnel working in low level radioactive contaminated areas. The arm sleeves may be worn in conjunction with part or all of the related radioactive contaminates protective outfit (hood, coveralls, gloves, and footwear items).

6.2 Ordering data Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When first article is required (see 3.2).
c. Selection of applicable levels of packaging (see 5.1).

6.3 Samples For access to samples address the procuring activity issuing the invitation for bids (see 3.1).

6.4 First article When a first article sample is required it shall be a preproduction sample and it shall be inspected and approved. The contracting officer should include specific instructions in acquisition documents regarding arrangements for inspections, quantity, and testing and approval of the first article.

Custodian:
Navy - NU

Review activity:
Navy - SH
DLA-CT

Preparing activity:
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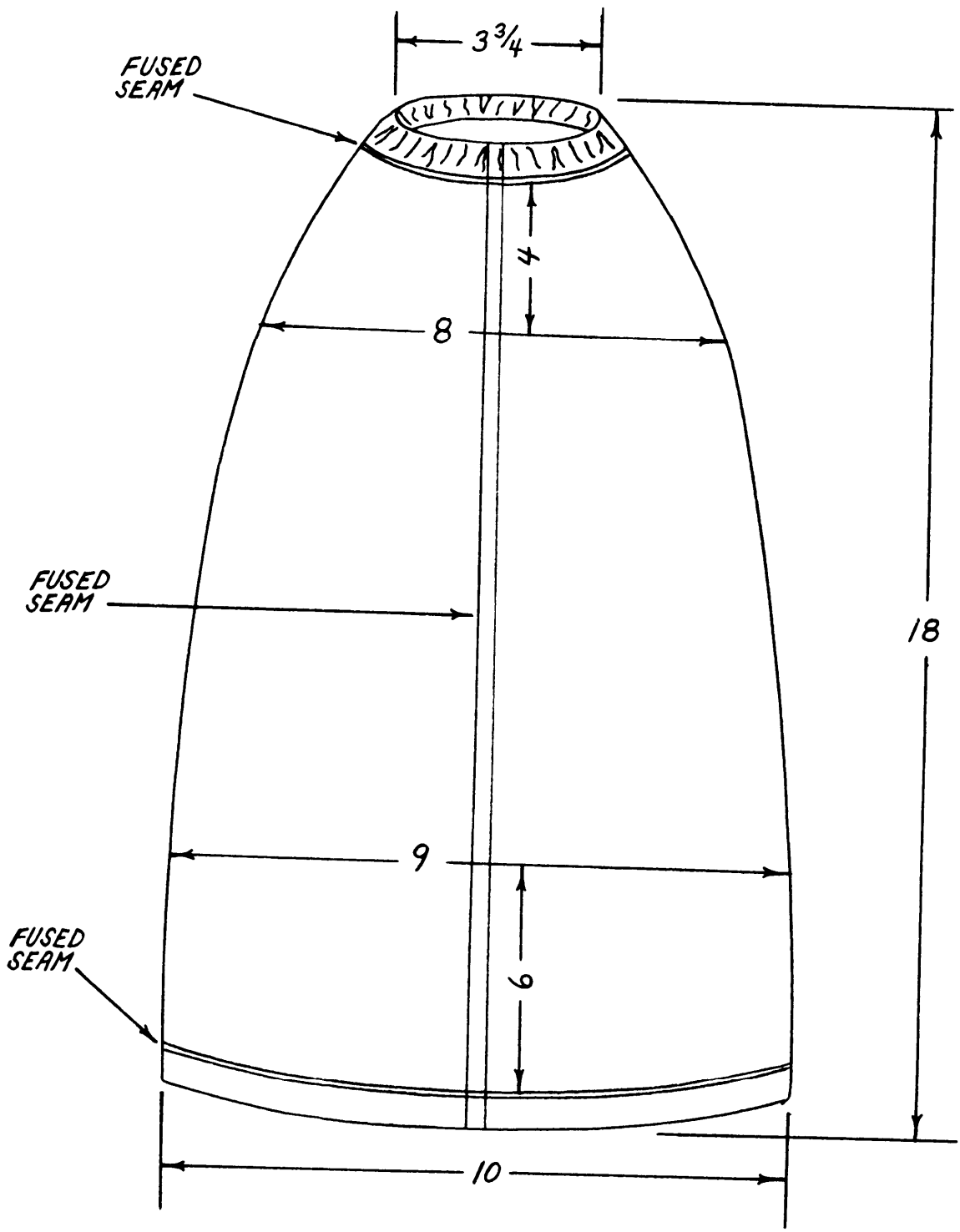


FIG. 1 - ARM SLEEVE, RADIOACTIVE CONTAMINANTS, PROTECTIVE

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-24914(NU)	ARM SLEEVE, RADIOACTIVE CONTAMINANTS, PROTECTIVE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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